



Remote sensing of diffuse attenuation coefficient using MODIS imagery of turbid coastal waters: A case study in Bohai Sea

Jun Chen^{a,b,*}, Tingwei Cui^c, Junwu Tang^d, Qingjun Song^e

^a School of Ocean Sciences, China University of Geosciences, Beijing 100083, China

^b Key Laboratory of Marine Hydrocarbon Resources and Environmental Geology, Qingdao Institute of Marine Geology, Qingdao 266071, China

^c First Institute of Oceanography, State Oceanic Administration, Qingdao 266071, China

^d National Ocean Technology Center, Tianjin 300111, China

^e National Satellite Ocean Application Service, Beijing 100081, China

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ABSTRACT

The objectives of this study are to evaluate the applicability of three existing retrieval algorithms of diffuse attenuation coefficients of downwelling irradiance at 490 nm, $K_d(490)$, for turbid Case II waters, and to improve these existing models using a simple semi-analytical (SSA) model. In this study, based on comparison of the $K_d(490)$ predicted by these models with field measurements taken in the Bohai Sea and Yellow Sea, it is shown that the SSA model provides a stronger performance than these three selected existing models. The atmospheric influences on the MODIS data are removed using an improved shortwave infrared band-based (ISWIR-based) model, which is capable of retrieving spectral remote sensing reflectance within 19% uncertainty. The $K_d(490)$ data was quantified from the MODIS images after atmospheric correction using the SSA model and Wang's model. The study results indicate that the SSA model produces 31.51% uncertainty in deriving $K_d(490)$ from MODIS data, which is 12.1% higher than Wang's model. This study demonstrates the potential of the SSA model in estimating $K_d(490)$ even in highly turbid coastal waters.

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1. Introduction

The light available in the water column at wavelength ranges from 400 to 700 nm in the visible parts of the spectrum (Bukata, Jerome, Kondratyev, & Pozdnyakov, 1995) is referred to as photosynthetically active radiation (PAR). It is a natural component of irradiance reaching the Earth's surface, and its potential to influence ecological processes and biogeochemical cycles in natural waters has long been recognized (Robert, Alexander, & Kirill, 1995). In aquatic environments, this light limits the growth of phytoplankton, and is a first order determinant of the response of phytoplankton to nutrient input in the sea. In other words, the PAR is used by phytoplankton for photosynthesis and constrains the type and distribution of algae species and benthic algae, thus contributing significantly to total primary production (Carter, Ross, Schiel, Howard-Williams, & Hayden, 2005). From the perspective of an energy cycle, the supply of PAR for phytoplankton growth in aquatic environments is a product of the input of solar radiation at the surface. The PAR supply is reduced by optically active constituents through absorption and scattering, which are increased by higher phytoplankton concentrations (Devlin et al., 2009). Therefore, light, together with nutrients, is a major governing parameter controlling production and photosynthetic processes in aquatic

environments (Lund-Hansen, 2004). Furthermore, the estimation of light diffuse attenuation coefficients, $K_d(\lambda)$, in an aquatic environment, is a critical factor to understand physical processes such as heat transfer in the upper layer of the ocean (Wu, Sathyendranath, & Platt, 2007).

Although capable of providing accurate measurements, traditional field monitoring on spatial distribution of $K_d(\lambda)$ is costly and time consuming, and thus confounds in coastal and oceanic systems by high temporal and spatial variability (Sweeney, Titterton, & Leonard, 1991). Recent advances in optical sensor technology have opened new opportunities to study biogeochemical processes in aquatic environments at spatial and temporal scales which previously had not been possible (Clavano, Boss, & Lee, 2007). These advances have allowed scientists to utilize ocean color satellite images to synoptically investigate large-scale surface features in the world's oceans (Chang & Gould, 2006). For example, since the launch of the Coastal Zone Color Scanner (CZCS) in 1978, the ocean-color community has provided maps of $K_d(\lambda)$ at large spatial scales, offering a great improvement in spatial and temporal resolution compared to in situ data (Barale, 1987). More recently, the launch of the Sea-viewing Wide Field-of-view Sensor (SeaWiFS) on September 18, 1997 has allowed the global ocean color data to be available to the science community in near-real time (Mueller & Trees, 1997). SeaWiFS was on a sun-synchronous satellite with a 2801 km swath width, providing 2-day coverage of the global ocean with a nadir resolution of ~1 km per pixel (Mueller & Fargion, 2002). Shortly afterwards,

* Corresponding author at: No.29, Xueyuan Road, Beijing, China.
E-mail address: cjun@cgs.cn (J. Chen).

two Moderate Resolution Imaging Spectroradiometers (MODIS) were launched by NASA on 1999 and 2002, respectively. These satellite sensors provided near-daily coverage of the global ocean. The study results obtained by Gordon and Voss (1999) indicated that MODIS was typically 2–3 times more sensitive than SeaWiFS, which in turn was approximately twice as sensitive as CZCS.

Traditionally, the in situ spectral $K_d(\lambda)$ had been measured by the ocean-color scientific community at 490 nm, following the primary studies performed in the 1970s (Smith & Baker, 1978). These achievements had a great impact on our knowledge of the world's oceans. Shortly afterwards, Austin and Petzold (1981) found an empirical relationship between CZCS observations and the diffuse attenuation coefficient for green light at 490 nm. Stumpf and Pennock (1991) used solutions obtained using the radiative transfer equation to derive relationships of water reflectance to $K_d(\lambda)$ in moderately turbid waters ($0.5 \text{ m}^{-1} < K_d(490) < 5 \text{ m}^{-1}$). Datasets collected from the National Oceanic and Atmospheric Administration (NOAA) Advanced Very High Resolution Radiometer (AVHRR) and field measurements obtained from Delaware Bay showed that the values of $K_d(490)$ calculated from the reflectance data agreed to within 60% of the observed values. However, due to limitations in the capabilities of the sensor and the atmospheric corrections, the results of these studies are suitable only for waters having an attenuation coefficient of less than 0.5 m^{-1} and a predominance of absorbing material (Siegel & Dickey, 1987). At present, the standard method (Mueller's model) to estimate values of $K_d(490)$ from remotely sensed data is through empirical relationships between $K_d(490)$ and the spectral ratio of water-leaving radiance at two wavelengths (Mueller & Trees, 1997). Such a model is insufficient to provide an understanding of the variation of $K_d(490)$, and involves large uncertainties inherent to empirical models (Lee, Du, & Arnone, 2005). In order to overcome these limitations, several empirical and semi-analytical models of $K_d(490)$ are developed for retrieving $K_d(490)$ maps from satellite data. Morel et al. (2007) proposed an empirical relationship between $K_d(490)$ and chlorophyll-*a* concentration (chl_a). However, due to the complete neglect of the influences of suspended sediment matters involved in the retrieval results, these results are only suitable for Case I waters.

In order to accurately derive $K_d(490)$ from turbid Case II waters, Lee et al. (2005, 2007) provided a semi-analytical relationship (Lee's model) between $K_d(490)$ and inherent optical properties based on the radiative transfer equation, with values of the model parameters derived from Monte Carlo simulations using the averaged particle phase function. The model was further tested with data simulated using significantly different particle phase functions, and the model-derived $K_d(490)$ was found to match the simulation results very accurately. The retrieval accuracy of this model is greatly dependent on the performance of the quasi-analytical algorithm (QAA model) suggested by Lee, Carder, and Arnone (2002). Ivey (2009) applied the QAA model to derive the respective $b_b(\lambda)$ from coastal waters of Puget Sound, Washington, Stocking Island, Bahamas, and West Florida Shelf, and indicated that the QAA model produces a poor performance in $b_b(\lambda)$ predictions, with an estimation uncertainty of >50%. Zhang, Stramski, and Reynolds (2010) also demonstrated that the QAA-derived spectral exponent of particulate backscattering generally leads to overestimation of both backscattering and absorption at wavelengths shorter than 551 nm, and may yield negative absorption coefficients in the Arctic waters as well as many coastal locations. Therefore, due to the failure of the QAA model in deriving inherent optical properties, Lee's model may produce poor performance in optically complex coastal waters.

Recently, Wang, Son, and Harding (2009) proposed a $K_d(490)$ model (referred to as Wang's model) for deriving $K_d(490)$ from coastal turbid waters. The model uses a relationship relating the backscattering coefficient at the wavelength 490 nm to the irradiance reflectance just beneath the surface, at the red wavelengths. Using the in situ-derived $b_b(490)$ relationship observed in Chesapeake Bay, Wang, Son, and Harding (2009) found that $K_d(490)$ models were formulated using the semi-analytical

approach. Match-up comparisons between the MODIS-derived and in-situ-measured $K_d(490)$ products observed in Chesapeake Bay indicate that the satellite-derived data using the proposed models are well correlated with the in situ measurements. In Wang's model, Wang, Son, and Harding (2009) related $b_b(490)$ to the irradiance reflectance just beneath the surface at wavelength 670 nm. However, in addition to backscattering by all particulate matter, the irradiance reflectance at 670 nm is also affected by total absorption coefficient (Gordon et al., 1988; Lee et al., 2002). If the total absorption coefficient at 670 nm varies between samples, the $b_b(490)$ model output may be different for the same $b_b(670)$. Thus, Wang's model may be violated in some complex cases, e.g. cases involving high chl_a concentrations. For the estimation of $K_d(\lambda)$, which is important for studies regarding heat budgets and photosynthesis (Ohlmann, Siegel, & Mobley, 2000; Sathyendranath, Prieur, & Morel, 1989; Wu et al., 2007), a model which is capable of providing a higher level of accuracy remains desired.

The objectives of this study are to validate the respective performances of Mueller, Lee and Wang's models, and to further improve these for turbid Case II waters using an innovative simple semi-analytical (SSA) model. The specific goals of the study are as follows: (1) to evaluate the accuracy of Mueller, Lee and Wang's models for accurately estimating $K_d(490)$ in turbid coastal waters; (2) to improve the performance of Mueller, Lee and Wang's models by a proposed SSA model with the spectral bands of the MODIS sensor; and (3) to compare the accuracy of Mueller, Lee and Wang's models, and the SSA model in estimating $K_d(490)$ from highly turbid and productive coastal waters in China.

2. Data, methods and techniques

2.1. Accuracy assessment

In this study, the Root-Mean-Square (RMS) of the ratio of modeled-to-measured values is used to assess the accuracy of atmospheric correction. This statistic will be referred to as *RMS* and is described by the following equation (Carder, Chen, Lee, Hawes, & Cannizzaro, 2003; O'Reilly et al., 1998):

$$RE_i = \left| \frac{x_{\text{mod},i} - x_{\text{obs},i}}{x_{\text{obs},i}} \right| \times 100\% \quad (1)$$

$$RMS = \sqrt{\frac{\sum_{i=1}^n RE_i^2}{n-2}} \times 100\% \quad (2)$$

where RE_i is the relative uncertainty of the *i*th observation, $x_{\text{mod},i}$ is the modeled value of the *i*th element, $x_{\text{obs},i}$ is the observed value of the *i*th element, and *n* is the number of elements.

2.2. Study area

The Bohai Sea (Fig. 1) is located between longitudes 117°38'E and 122°12'E, and latitudes 37°10'N and 40°51'N, and has a total area of $7.8 \times 10^4 \text{ km}^2$. Surrounded by the land on three sides, the Bohai Sea is semi-enclosed and usually divided into four sections, namely Liaodong Bay, Bohai Bay, Laizhou Bay and the Central Bohai Sea. The Bohai Sea is very shallow, with an average water depth of 18 m, and the deepest areas (~80 m) are located in the Bohai Strait (Cui et al., 2010), which is the link between the Bohai Sea and the Yellow Sea. Several large rivers pour into the Bohai Sea, carrying a large amount of organic and inorganic suspended matter, with an average annual run-off of $5.8 \times 10^{10} \text{ m}^3$ and annual sediment transport of $1.1 \times 10^9 \text{ tons}$ (Chen, Quan, Wen, & Cui, 2013). As the second largest river in the world in terms of sediment loadings, the Yellow River has discharged approximately 10^9 tons/year of sediments into the Bohai Sea, throughout the past several thousand

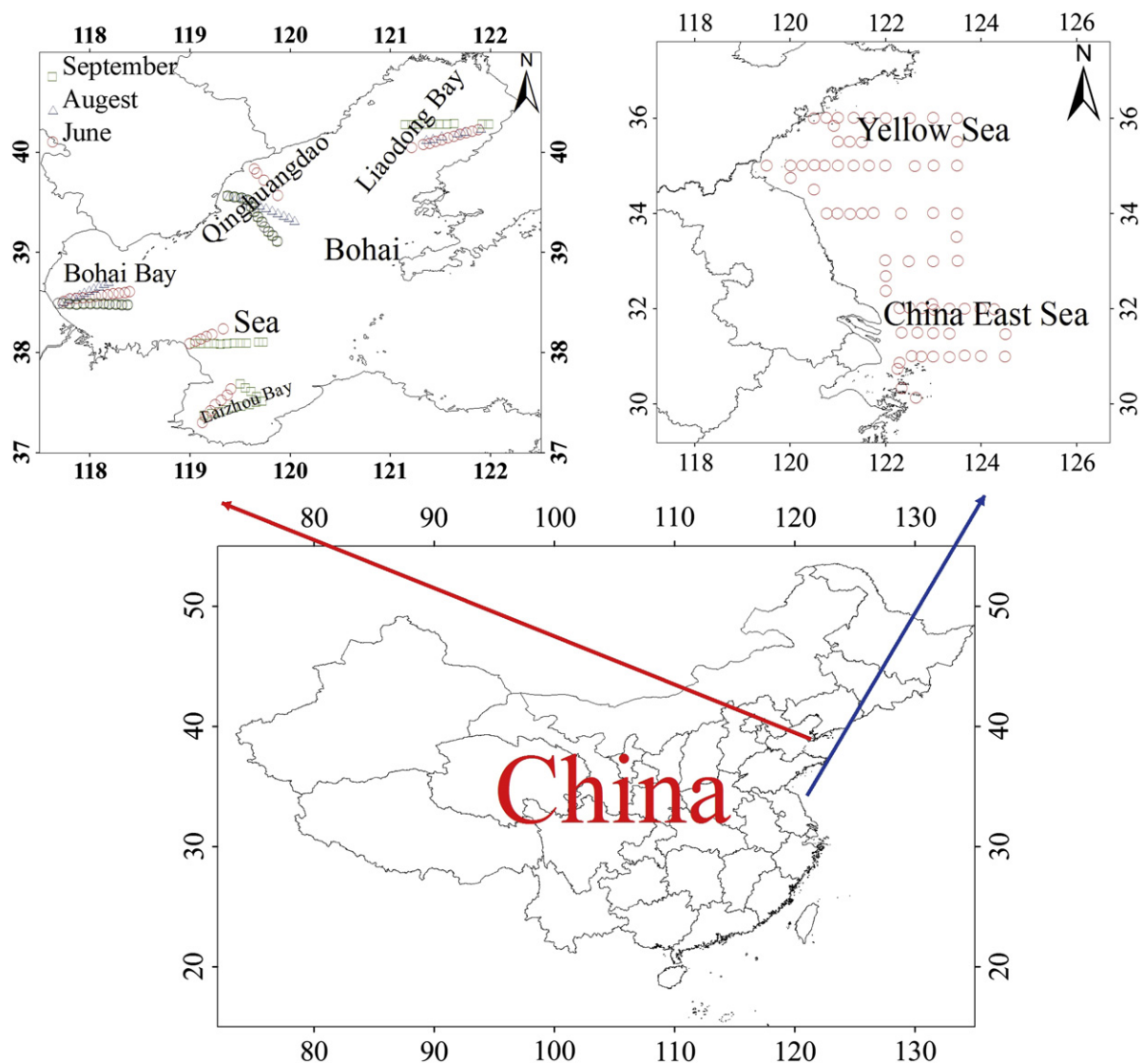


Fig. 1. Study area and experimental stations.

years (Chen & Quan, 2013). This clearly indicates that chlorophyll-*a* is not the only contributor of water optical properties, and that the majority of the Bohai Sea water falls into the Case II category (Morel, 1997).

River discharge, re-suspension and exchange with water masses of the Yellow Sea are the factors determining the spatial distribution of suspended sediment matters in the Bohai Sea. China's Yellow Sea (Fig. 1), located between longitudes 120°11'E and 126°14'E, and latitudes 30°16'N and 36°54'N, is one of the largest marginal seas in the world, and is partially enclosed by coastlines. Due to the large rivers flowing into the Yellow Sea which carry vast amounts of mineral-rich soil, the water of the sea actually appears yellow. The Yellow Sea is also contaminated by industrial pollution, agricultural runoff and domestic sewage (Zhang, Tang, Dong, & Song, 2010). Thus, the water mass of the Yellow Sea is classified as typical Case II water.

2.3. Datasets used and field measurements

In order to assess the applicability of Mueller, Lee and Wang's models and the SSA model in deriving $K_d(490)$ from MODIS data in turbid coastal waters, three independent datasets (Fig. 1) consisting of simultaneous measurements of above-water remote sensing reflectance and $K_d(\lambda)$ were used. These datasets were obtained over two different shelf seas during 26 independent cruises performed in the Bohai and Yellow

Seas. The first dataset was used for model calibration, while the second and third were used for model validation. The first dataset (Fig. 2A and B), containing 84 samples, was taken in August and September 2005 in the Bohai Sea; the second dataset (Fig. 2C and D), containing 42 samples, was collected in June 2005 in the Bohai Sea; and the third dataset (Fig. 2E), containing 69 samples, was collected from the Yellow Sea during the cruises carried out in September 2003.

Above-surface optical measurements were made by means of a field spectroradiometer (Spectral Devices, Boulder, CO, ASD), simultaneous with in-water optical measurements, according to NASA ocean-optics protocols (Mueller & Fargion, 2002). The performance of the $K_d(\lambda)$ retrieval model depends on several factors, including sufficient stations representing different conditions in the study area, and careful data quality control of in situ measurements. The reflectance measured by ASD may have involved errors caused by instrument calibration, ocean conditions (sea surface roughness and wind speed), and data processing methods. It is important to inspect the data quality carefully and to remove inaccurate measurements of the reflectance before the data are analyzed. In order to improve the accuracy of the field measurements, the spectroradiometer had been calibrated by the manufacturer on an annual basis. Various measurements (three repeated measurements recorded over a short period of time) were made at each station in order to estimate the precision (RMS values of three repeated

measurements) associated with each measurement, and the average measurements (mean values of three repeated measurements) with <5% RMS at each station were selected for model calibration and validation.

In situ downwelling irradiance measurements must be collected following a community vetted protocol (Werdell & Bailey, 2005), in order to avoid ship shadow and reflection. The measurements were performed from the sunny side of the ship, and were usually taken from 0.6 to 6 m, at 0.3 m sampling intervals. The profile of downwelling irradiance, $E_d(\lambda)$, was measured by means of a TACCS radiometer (Satlantic). At each station, the TACCS instrument was set to sample for 3 min, at a sampling rate of 2 samples per second (for a total of 90 measurements at each station). The natural logarithm of the measured $E_d(\lambda)$ was plotted against depth, and an estimate of $K_d(\lambda)$ was acquired from the resulting

slope (Pierson, Kratzer, Strömbeck, & Håkansson, 2008). The measurements obtained in this study suggest that the first optical depth in the Bohai Sea usually ranges between 0.6 and 5 m and rarely exceeds 6 m. This indicates that in this study the measurements of underwater irradiance were made to a sufficient depth, in order to accurately estimate the $K_d(\lambda)$. The total absorption coefficient minus pure water absorption coefficient, denoted as $a_{t-w}(\lambda)$, is measured by a Wetlabs AC9 photometer. The temperature salinity and reflective tube scattering corrections were performed after the cruise. The correction method is described in WET Labs, Inc. (2008). The data for $a_{cdm}(\lambda)$ were measured according to the SeaWiFS protocols (Mueller & Fargion, 2002).

The total suspended matter (TSM) concentration was measured by a weighing method proposed by Mueller and Fargion (2002). The water sample was filtered with 450 nm filter (Whatman GF/F filters)

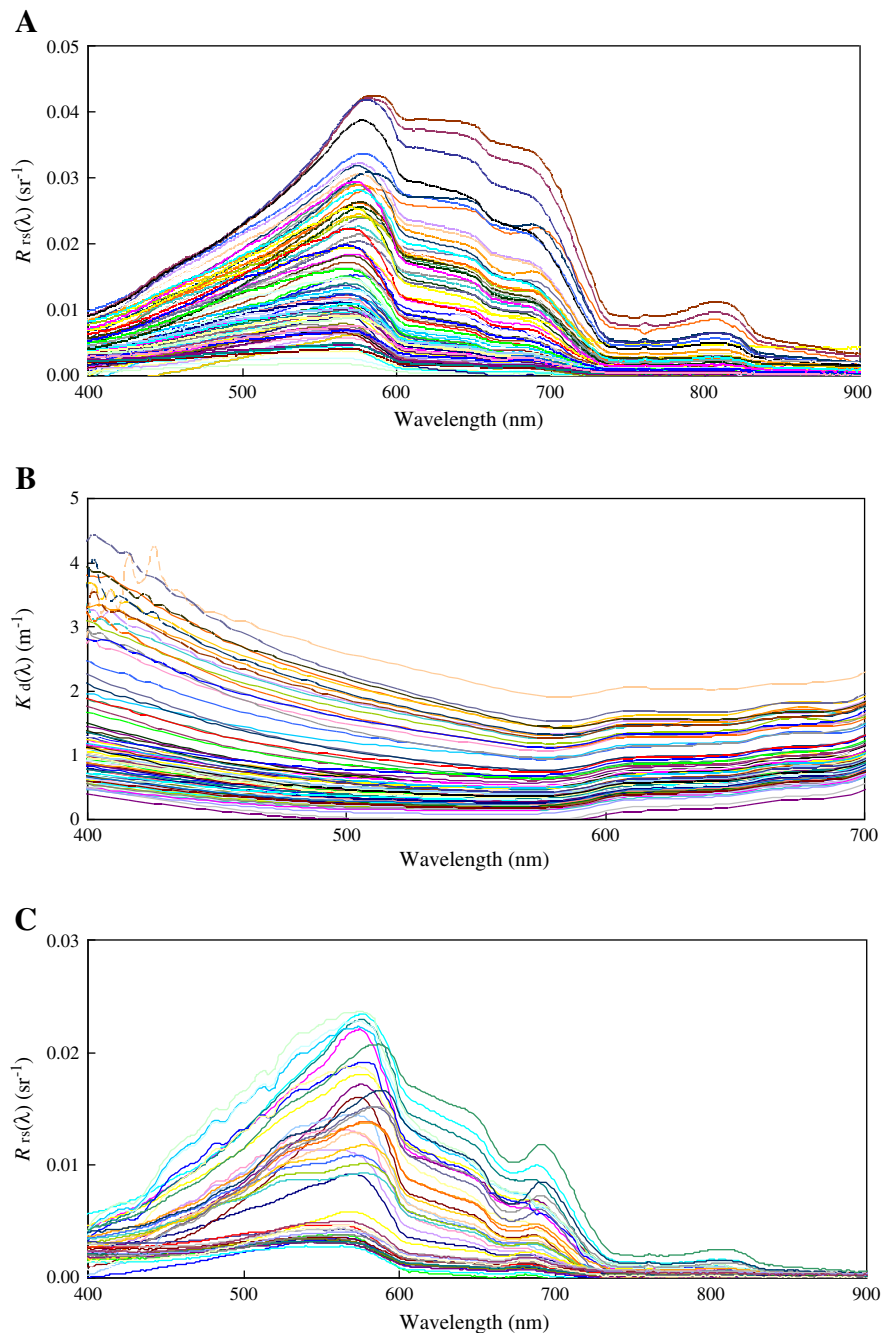


Fig. 2. Spectral curves of $R_{rs}(\lambda)$ and $K_d(\lambda)$ of three independent dataset. (A) and (B) The first dataset collected from Bohai Sea in August and September 2005; (C) and (D) the second dataset collected from Bohai Sea in June 2005; and (E) the third dataset collected from Yellow Sea in 2003.

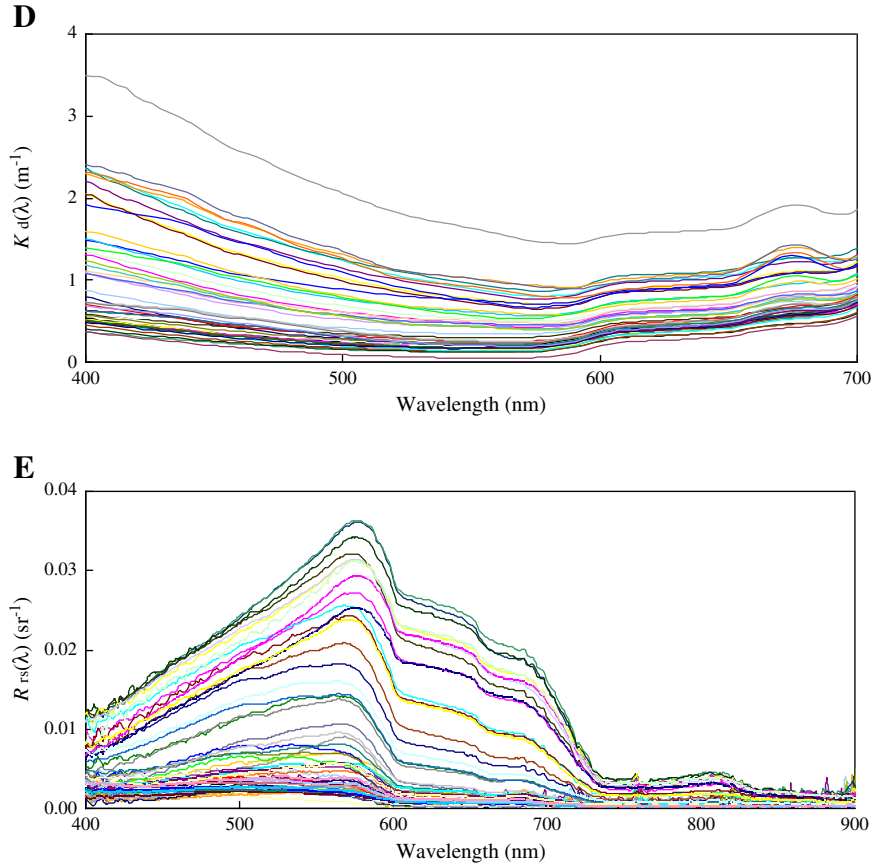


Fig. 2 (continued).

and a vacuum filtration system. The filter pad was flushed three times with 0.00005 m³ distilled water, in order to flush away the salt. The dry-weight of the filter-pad was weighed using an electronic analytic scale. The blank filter and sampled filter-pad were weighed until the difference between two successive TSM concentrations calculated from the scale reading was within 0.01 mg/l (Zhang, Tang, et al., 2010). Chla concentration was measured by passing up to 500 ml of seawater through Whatman G/F glass filters. These filters were extracted overnight in 90% acetone neutralized in a refrigerator using NaHCO₃, without light, as described by Devlin et al. (2009). After centrifugation, extract fluorescence was measured using a Turner Designs Model 10-AU-005 CE fluorometer, before and after acidification, with 8% HCl. This was used to calculate concentrations of the chl a. The fluorometer had been calibrated using pure solutions of chl a (Sigma Chemical Co, ORGIN) with concentration determined spectrophotometrically. Replicate measurements of chl a had typical standard errors of approximately 10% RMS.

2.4. Construction of SSA model

According to the classical equation of transfer, K_d may be written as (Appendix A):

$$K_d(490) = \frac{1 + \mu_d p s(490)}{\mu_d [1 - \mu_u p s(490)]} a(490) \quad (3)$$

where s is the ratio of backscattering to absorption, p represents the ratio of r_d to the sum value of μ_u and μ_d , r_d represents the mean upward scattering coefficient of the downward traveling photons, and μ_u and μ_d are the upward and downward average cosines, respectively. Some examples of the values of μ_u , μ_d , and r_d can be found from the study results of Aas (1996). Due to the fact that satellites and many other sensors measure remote sensing reflectance from above the water surface, it is

necessary to convert above-surface remote sensing reflectance spectral $R_{rs}(\lambda)$ into below-surface spectral $r_{rs}(\lambda)$ (Gordon et al., 1988; Lee, Carder, & Du, 2004):

$$r_{rs}(\lambda) = \frac{R_{rs}(\lambda)}{T + \kappa Q R_{rs}(\lambda)} \quad (4)$$

where $T = t^u t^d / n^2$ with t^u the radiance transmittance from below to above the surface and t^d the irradiance transmittance from above to below the surface; n is the refractive index of waters; κ is the water-to-air internal reflection coefficient; and Q is the ratio of upwelling irradiance to upwelling radiance evaluated below the surface. In general, a nadir-viewing sensor and remote sensing domain, Q , ranges between 3 and 6 (Morel & Gentili, 1996). As $R_{rs}(\lambda)$ is small in comparison to most oceanic and coastal waters, the variation of the Q value has only little influence on the conversion between $R_{rs}(\lambda)$ and $r_{rs}(\lambda)$ (Lee et al., 2002). From calculated HYDROLIGHT $R_{rs}(\lambda)$ and $r_{rs}(\lambda)$ values, it has been found that $T \approx 0.52$ and $\kappa Q \approx 1.7$ for optically deep waters and a nadir-viewing sensor (Mobley, 2008).

The inherent optical properties inversion model assumes that the relationship between $r_{rs}(\lambda)$ and the ratio of absorption ($a(\lambda)$) to backscattering ($b_b(\lambda)$) is known and stable, for example (Gordon, Brown, & Jacobs, 1975):

$$r_{rs}(\lambda) = \frac{s(\lambda)}{1 + s(\lambda)} \left[l_0 + l_1 \frac{s(\lambda)}{1 + s(\lambda)} \right] \quad (5)$$

where l_0 and l_1 are empirical coefficients. Gordon et al. (1988) found that $l_0 \approx 0.0949$ and $l_1 \approx 0.0794$ for Case I waters, while Lee, Carder, Mobley, Steward, and Patch (1999) suggested that l_0 of 0.084 and l_1 of 0.17 are more suitable for higher scattering coastal waters. In this study, Lee's results are used, due to the fact that the Bohai Sea is classified as a

turbid Case II water. Therefore, knowing $r_{rs}(\lambda)$, $s(\lambda)$ may be quickly calculated using Eq. (5):

$$s(\lambda) = \frac{b_b(\lambda)}{a(\lambda)} = \frac{2r_{rs}(\lambda)}{I_0 + \sqrt{I_0^2 + 4I_1 r_{rs}(\lambda) - 2r_{rs}(\lambda)}}. \quad (6)$$

The backscattering coefficient may generally be expressed as a function of the spectral irradiance reflectance just beneath the sea surface and the absorption (Gordon et al., 1988). Doron, Babin, and Hemblise (2007) showed that the backscattering coefficient at the wavelength 490 nm, $b_b(490)$, may be linearly related to the irradiance reflectance just beneath the surface at the wavelength 709 nm. Based on this concept, Wang, Son, and Harding (2009) related $b_b(490)$ to the irradiance reflectance just beneath surface at the wavelength 670 nm as follows:

$$b_b(490) = C_0 + C_1 R(670) \quad (7)$$

where C_0 and C_1 are the empirical coefficients, and $R(670)$ refers to the irradiance reflectance just beneath the surface at wavelength 670 nm. Doron et al. (2007) proposed that C_1 can be determined using the formula of $[b_{bp}(490)a(670)] / [b_{bp}(670)f(670)]$, where $f(\lambda)$ is the coefficient that relates ocean upwelling irradiance reflectance to the ocean inherent optical properties, and $b_{bp}(\lambda)$ refers to backscattering coefficient associated with suspended particles. Wang, Son, and Harding (2009) found that $C_0 = 7 \times 10^{-4}$ and $C_1 = 2.7135$ were reasonable for Case II waters. Traditionally, $b_{bp}(490)$ is wavelength dependence of $b_{bp}(670)$ (Chen, Yao, & Quan, 2013; Gallie & Murtha, 1992; Zawada et al., 2007), so the ratio of $b_{bp}(490)$ to $b_{bp}(670)$ can be approximated to a “constant” that depends heavily on the size and optical properties of suspended particles (Bowers, Braithwaite, Nimmo-Smith, & Graham, 2009). If the influences of $f(\lambda)$ associated with the different geometry of the light field emerging from the water body can be ignored (Carder et al., 2003; Gordon et al., 1988), the value of C_1 is still affected by $a(670)$. In turbid coastal waters, the reflectance was highly variable and several spectra characteristics exist at around 670 nm, e.g. an absorption peak at around 674 nm corresponds to the chl *a* absorption maximum in the red band (Ahn & Shanmugam, 2006). In other words, the absorption due to optically active constituents may vary among samples, thus the value of C_1 would be different for the same $b_{bp}(490)$ and $b_{bp}(670)$. In addition, Tzortziou et al. (2006) showed that the absorption of particulate matter at around 670 nm could not be neglected in Chesapeake Bay; and Tassan and Ferrari (2003) found that the aquatic particle absorption varies significantly depending on the type of suspended particles. Fig. 3 shows the histogram results in $a(667)$ measured from the Bohai Sea, and that the $a(667)$ varies from

0.4409 to 0.8224 m^{-1} , the average and SD being 0.5131 m^{-1} and 0.0566 m^{-1} , and the CV being 11.03% for the Bohai Sea. This is to say that it may result in >10% uncertainty in determination of $b_b(490)$ using Eq. (7), if the method proposed by Wang, Son, and Harding (2009) is used to initialize the value of C_1 , and thereby may affect the performance of Wang's $K_d(490)$ model in optical complex waters. In order to account for this, a two-band model is developed for assessing $a(490)$ in turbid coastal waters. The spectral slope and bias for total absorption minus water absorption is defined as follows (Chen, Yao, et al., 2013; Tzortziou et al., 2007):

$$a_{t-w}(667) = \varepsilon_{aa} a_{t-w}(490) + \varepsilon_{ab}. \quad (8)$$

The analogous expression for total backscattering is defined as follows (Chen, Yao, et al., 2013; Lee et al., 2002; Smyth, Moore, Hirata, & Aiken, 2006):

$$\varepsilon_b = \frac{b_b(490)}{b_b(667)} \quad (9)$$

where ε_{aa} and ε_{ab} are spectral slope and bias for $a_{t-w}(\lambda)$ between 490 and 667 nm; and ε_b refers to the spectral slope for $b_b(\lambda)$ between 490 and 667 nm. Fig. 4 shows the spectral relationship of $a_{t-w}(488)$ vs. $a_{t-w}(676)$ and $b_b(488)$ vs. $b_b(676)$ in the Bohai Sea, indicating that the value for ε_{aa} , ε_{ab} , and ε_b converged to a narrow range of value for all the components with $\varepsilon_{aa} = 0.2513$, $\varepsilon_{ab} = 0.04$, and $\varepsilon_b = 0.9207$, respectively, and the corresponding determination coefficients are 0.876 and 0.971, respectively. These findings imply that the values for ε_{aa} , ε_{ab} , and ε_b may be approximated to the unknown constants. Substituting Eqs. (6) and (9) into the difference $b_b(667) [s^{-1}(667) - \varepsilon_{aa}\varepsilon_b s^{-1}(490)]$, yields

$$b_b(667) = \frac{a_{t-w}(667) + a_w(667) - \varepsilon_{aa} a_{t-w}(490) - \varepsilon_{aa} a_w(490)}{s^{-1}(667) - \varepsilon_{aa}\varepsilon_b s^{-1}(490)}. \quad (10)$$

By inserting Eq. (8) into Eq. (10), the $b_b(667)$ can be calculated as follows:

$$b_b(667) = \frac{a_w(667) - \varepsilon_{aa} a_w(490) + \varepsilon_{ab}}{s^{-1}(667) - \varepsilon_{aa}\varepsilon_b s^{-1}(490)}. \quad (11)$$

Provided that $b_b(667)$ is known, $a(667)$ can be determined from the ratio of $b_b(667)/s(667)$:

$$a(667) = \frac{a_w(667) - \varepsilon_{aa} a_w(490) + \varepsilon_{ab}}{1 - \varepsilon_{aa}\varepsilon_b q} \quad (12)$$

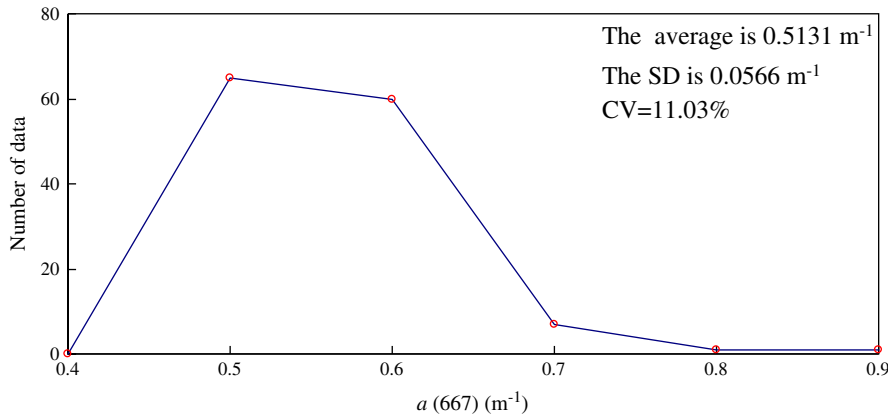


Fig. 3. Histogram results in $a(667)$ measured from Bohai Sea, where SD represents the standard derivation, and CV refers to the ratio of SD to average of parameter.

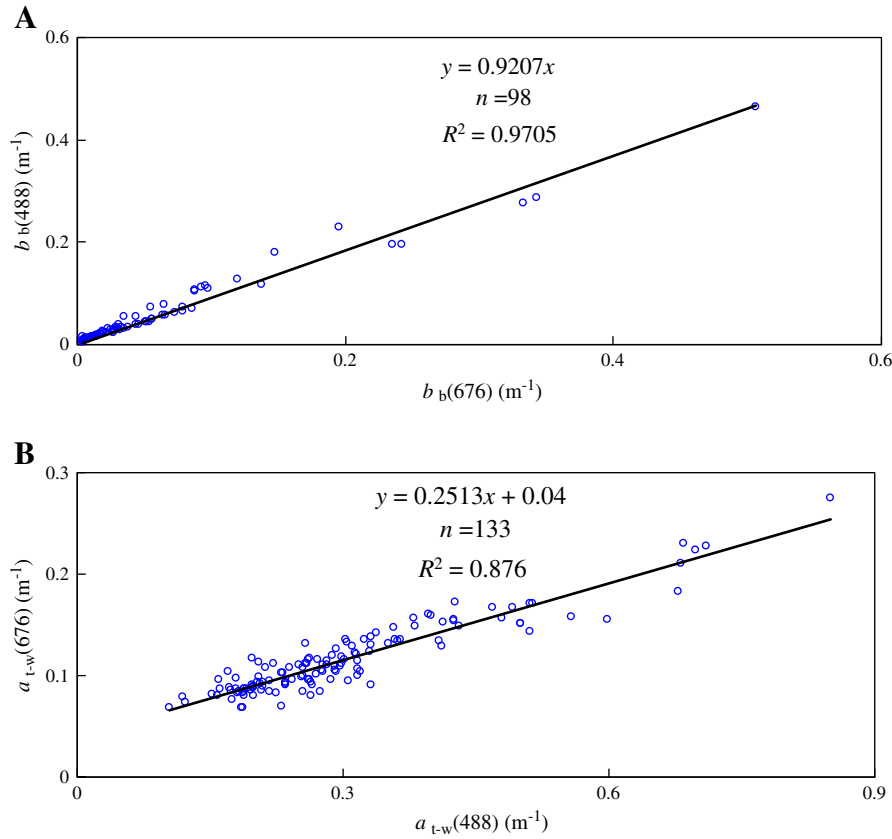


Fig. 4. Spectral relationship of inherent optical properties in Bohai Sea. (A) $b_b(488)$ vs. $b_b(676)$; and (B) $a_{t-w}(488)$ vs. $a_{t-w}(676)$.

where q represents the ratio of $s(667)$ to $s(490)$. Inserting Eq. (8) into Eq. (12) yields the $a(490)$ formula as follows:

$$a(490) = \frac{a_w(667) - \varepsilon_{aa}a_w(490) + \varepsilon_{ab}}{\varepsilon_{aa} - \varepsilon_{aa}^2\varepsilon_bq} + \frac{\varepsilon_{aa}a_w(490) - a_w(667) - \varepsilon_{ab}}{\varepsilon_{aa}}. \quad (13)$$

Inserting Eq. (13) into Eq. (3) yields the following:

$$K_d = \frac{1 + \mu_dps}{(1 - \mu_u ps)} \left(\frac{h_0}{1 - \varepsilon_{aa}\varepsilon_bq} + h_1 \right) \quad (14a)$$

$$h_0 = \frac{a_w(667) - \varepsilon_{aa}a_w(490) + \varepsilon_{ab}}{\mu_d\varepsilon_{aa}} \quad (14b)$$

$$h_1 = \frac{\varepsilon_{aa}a_w(490) - a_w(667) - \varepsilon_{ab}}{\mu_d\varepsilon_{aa}} \quad (14c)$$

where K_d and $a(490)$ are both greater than zero. Because $s < 0.4$, $q < 1.8$, and $\varepsilon_{aa}\varepsilon_b < 0.25$ found in the Bohai Sea in this case study, both $\mu_u ps$ and $\varepsilon_{aa}\varepsilon_bq$ must be greater than 0 but smaller than 1. Therefore, according to the summation formula of geometric progression (Chen, Yu, & Jin, 2003), it was found that Eq. (13) can be expanded in a series, as follows:

$$K_d(490) = (1 + \mu_dps) \sum_{i=0}^{\infty} (\mu_u ps)^i \left[h_0 \sum_{j=0}^{\infty} (\varepsilon_{aa}\varepsilon_bq)^j + h_1 \right]. \quad (15)$$

Based on the second order approximation of geometric progression using the intermediate value theorem (Chen et al., 2003), the SSA model may be approximately expressed as follows:

$$K_d(490) \approx \left[1 + (\mu_d p + m_0 p)s + \mu_d p m_0 s^2 \right] (h_0 + h_1 + \varepsilon_{aa}\varepsilon_bq + v_0 q^2) \quad (16)$$

where m_0 and v_0 are the intermediate constants of geometric progression on $\mu_u ps$ and $\varepsilon_{aa}\varepsilon_bq$, respectively.

3. Results

3.1. Constituent concentrations

The dataset encompassed varying optical conditions and included a wide range of phytoplankton taxonomic groups including diatoms, dinoflagellates, nanophytoplankton and picophytoplankton (Cui et al., 2010). In each of the datasets taken in the Bohai Sea in 2005, the concentration of chl *a* and TSM, as well as $K_d(490)$, varied over one order of magnitude, except for TSM, which varied by two orders of magnitude (e.g. 0.118–4.117 m^{-1} $K_d(490)$, 1.02–8.32 $\mu g/l$ chl *a* concentration, 0.96–46.8 mg/l and 2.6–62.4 mg/l TSM concentration) (Table 1). In

Table 1

Descriptive statistics of the bio-optical parameters measured: chl *a* concentration, $K_d(490)$, and TSM concentration (SD refers to standard deviation and $a_{cdom}(490)$ denotes the absorption by CDOM at 490 nm).

	Min	Max	Median	Average	SD
<i>a. Calibration dataset taken from Bohai Sea in September 2005, 84 samples</i>					
$K_d(490)$, 1/m	0.173	4.068	0.598	0.907	0.802
TSM, mg/l	0.960	46.800	6.600	10.290	7.438
chl <i>a</i> , $\mu g/l$	1.271	5.603	2.712	2.945	0.943
$a_{cdom}(490)$, 1/m	0.008	1.100	0.158	0.213	0.191
<i>b. The first validation dataset taken from Bohai Sea in June 2005, 42 samples</i>					
$K_d(490)$, 1/m	0.189	3.646	0.598	0.743	0.634
TSM, mg/l	2.600	62.400	8.300	15.852	17.950
chl <i>a</i> , $\mu g/l$	0.945	9.832	2.749	3.045	1.883
$a_{cdom}(490)$, 1/m	0.008	1.482	0.216	0.302	0.264
<i>c. The second validation dataset taken from Yellow Sea April and September 2003, 69 samples</i>					
$K_d(490)$, 1/m	0.041	1.945	0.262	0.460	0.522

the Bohai Sea, the $K_d(490)$, chl *a* and TSM concentrations vary depending on the season. In September the $K_d(490)$ varies from 0.178 to 4.117 m^{-1} , while the chl *a* and TSM concentrations vary from 1.271 to 4.718 $\mu\text{g/l}$ and 0.96 to 46.8 mg/l , respectively. In August the variation ranges of $K_d(490)$ and TSM concentration were smaller, half the size of those in September (denoted in Table 1 using SD). In June the $K_d(490)$ was smaller than that in September but larger than that in August, while the TSM concentration was larger than in these latter two months.

As in the case of the Bohai Sea in June–September 2005, $K_d(490)$ vs. chl *a* concentration, $K_d(490)$ vs. TSM concentration, and $K_d(490)$ vs. $a_{\text{cdom}}(490)$ in these datasets (Fig. 2 and Table 1) were not significantly correlated, where $a_{\text{cdom}}(490)$ refers to the absorption due to colored dissolved organic matter (CDOM) at 490 nm. However, when we ignored outliers with TSM > 20 mg/l , the performances of $K_d(490)$ depended on TSM concentration increased significantly. Similarly, when we ignored

outliers $K_d(490) > 1.5 \text{ m}^{-1}$, the performance of $K_d(490)$ depended on chl *a* concentration significantly increased. These findings imply that the optical properties of Bohai Sea waters are very complex, i.e., some Bohai Sea waters are TSM dominated types while some others are chl *a* dominated types. This is consistent with the results from Jiang, Pohlmann, Sun, and Starke (2004) and Cui et al. (2010). For all the datasets, the coefficients of determination (R^2) for a linear relationship between $K_d(490)$ and chl *a* concentration, $K_d(490)$ and TSM concentration, and $K_d(490)$ and $a_{\text{cdom}}(490)$ were smaller than 0.2 (Fig. 5). The lack of a significant relationship between $K_d(490)$ and chl *a* concentration, $K_d(490)$ and TSM concentration, and $K_d(490)$ and $a_{\text{cdom}}(490)$ indicated that each parameter was independently controlling the optical properties of these waters, thereby confirming that the Bohai Sea water falls into the Case II category. This is consistent with the results obtained by Cui et al. (2010).

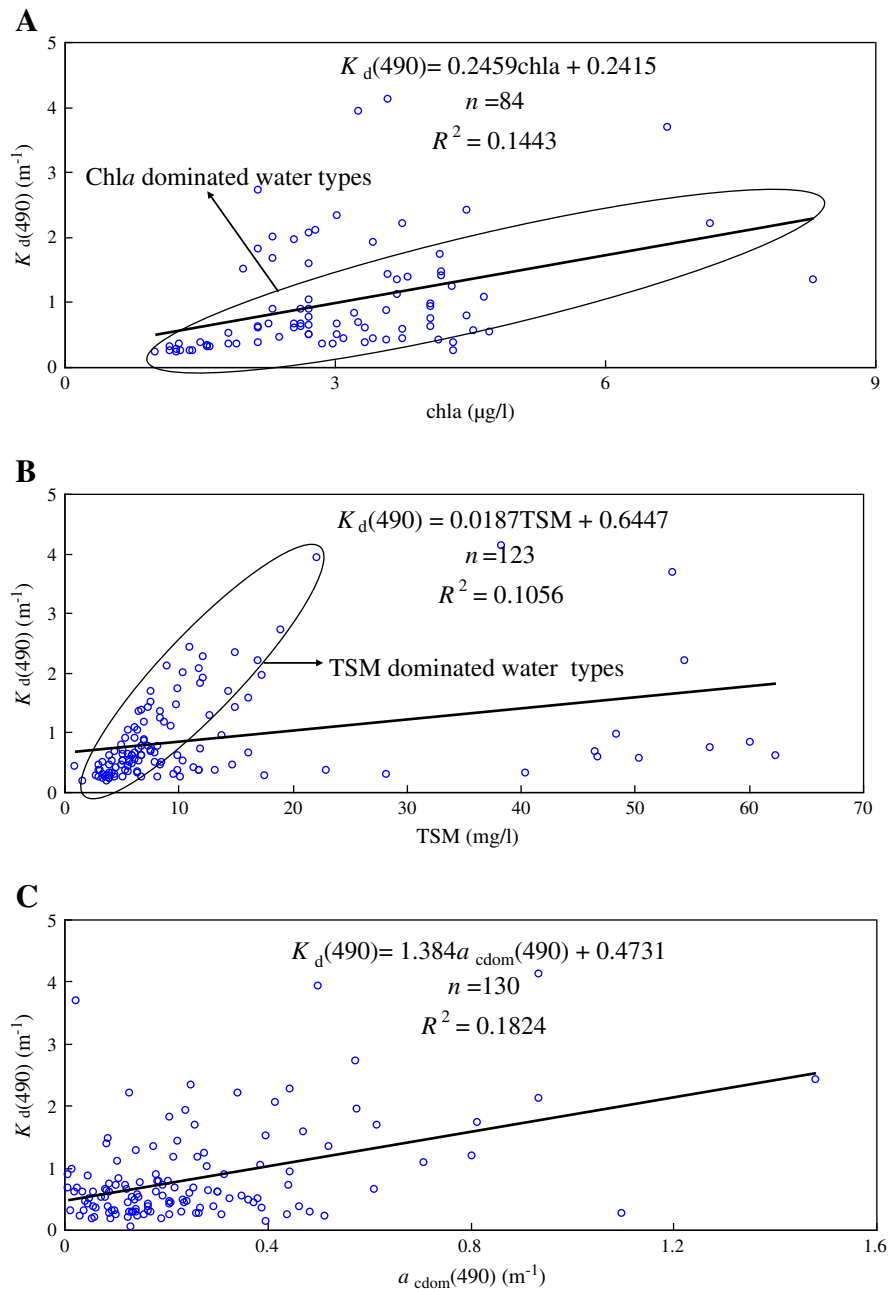


Fig. 5. Scatter plots of $K_d(490)$ versus water quality parameters. (A), (B), and (C) are the relationships of $K_d(490)$ vs. chl *a* concentration, $K_d(490)$ vs. TSM concentration, and $K_d(490)$ vs. $a_{\text{cdom}}(490)$ in Bohai Sea.

Table 2 shows a comparison of the averaged suspended particle size and suspended sediment concentration among samples collected from Laizhou Bay, Bohai Bay, Liaodong Bay, and Qinhuangdao Bay. The highest TSM concentration was found in Laizhou Bay (17.21 mg/l), mostly due to the Yellow River Estuary, which is an important TSM contributor to the Bohai Sea. The finest suspended particles were found in Bohai Bay. Fig. 6 shows that the averaged remote sensing reflectance in Bohai Bay is slightly larger than that in Laizhou Bay, but the TSM concentration and particle diameter (d) in the former are averaged 3.48 mg/l and 1.338 μm lower than the latter (Table 2). Bowers and Binding (2006) indicated that smaller sized sediments generally lead to a higher spectral reflectance. Therefore, particle size is an important contributor to the water optical properties in the Bohai Sea. Furthermore, the optical properties in the Bohai Sea are very complex, not only due to the large amount of terrestrial organic and inorganic suspended matter found there, but also because of the different particle size in spatial distribution.

3.2. Spectral characteristics

As shown in Fig. 2, the spectra curves for the three datasets were quite similar in shape to the typical reflectance spectra collected in highly turbid waters (Chen, Fu, & Sun, 2012; Cui et al., 2010; Gitelson et al., 2008). The remote sensing reflectance in the blue range (400–500 nm) was very low. In Bohai Bay, the averaged remote sensing reflectance was $<0.013 \text{ sr}^{-1}$, which is slightly larger than that in Laizhou Bay, but slightly smaller than that in the Bohai Sea (Fig. 6). In Liaodong Bay the blue reflectance was even lower ($<0.007 \text{ sr}^{-1}$), approximately two times larger than that of Qinhuangdao Bay. Reflectance at this spectral region did not have pronounced spectral features for the broad range of TSM (0.96–62.4 mg/l) and chl a concentration (1.02–8.32 $\mu\text{g/l}$) at all sites (Table 1). A reflectance trough at around 440 nm, corresponding to the chl a absorption peak (Gons, Auer, & Effler, 2008), was indistinct for the dataset used in this study, due to the low chl a concentration ($<8.32 \mu\text{g/l}$), and only have been distinct when the chl a concentration was $>150 \mu\text{g/l}$ (Gitelson et al., 2008).

Reflectance in the green region (500–600 nm) was much higher than in the blue and red ranges. Many factors contribute to the reflectance features in the blue and green ranges, which are primarily representative of absorption by dissolved organic matter and tripton, as well as backscattering caused by suspended particles (Mobley, 1994). In the red range (600–700 nm), the reflectance was highly variable, and several spectral characteristics existing in this region. A second minimum at around 674 nm corresponds to the red chl a absorption maximum (Ahn & Shanmugam, 2006). A distinct peak was located between 690 and 710 nm. This peak is the result of both high backscattering and a minimum in absorption by all optically active constituents, including pure water (Gitelson & Merzlyak, 1994). The reflectance in the near-

Table 2

Comparison of averaged suspended particle size and suspended sediment concentration between samples collected from Laizhou Bay, Bohai Bay, Liaodong Bay, and Qinhuangdao Bay.

	Laizhou Bay	Bohai Bay	Liaodong Bay	Qinhuangdao Bay
d , μm	7.750	6.412	7.218	20.98
TSM, mg/l	17.21	13.73	7.993	6.599

infrared (NIR) range (700–900 nm) varied widely. The spectral curves ranging from 700 to 800 nm were very flat, due to the low backscattering and absorption caused by optically active constituents (Robert et al., 1995), with the exception of pure waters. A third distinct peak was located at around 814 nm, due to the high backscattering suspended particles, while the TSM was $>10 \text{ mg/l}$. Due to the high absorption caused by pure waters, the reflectance values of this peak are usually lower than those at around 570 and 700 nm.

3.3. Model calibration, evaluation and comparison

Due to the fact that the MODIS is a broad-band scanning radiometer, the $R_{rs}(\lambda)$ measured by ASD must be weighted by the MODIS sensor spectral response functions before model calibration and evaluation is performed.

3.3.1. Evaluation of Mueller, Lee and Wang's models

Mueller, Lee and Wang's models have been described in detail in various references (Lee et al., 2005; Mueller & Trees, 1997; Wang, Son, & Harding, 2009). The coefficients of these three models were adjusted according to the bio-optical information collected from the Bohai Sea (Table 3). The models' evaluation was based on comparison of the $K_d(490)$ predicted by these models with $K_d(490)$, as shown in Fig. 2A and B. Fig. 7 shows the comparison between the model-derived and field measured $K_d(490)$. It was found that both Mueller and Lee's $K_d(490)$ estimations were accurate for clear waters ($K_d(490) < 0.8 \text{ m}^{-1}$). However, above 0.8 m^{-1} , both models produced poor performance in deriving $K_d(490)$ from the Bohai Sea. For example, when using Mueller's model, the $K_d(490)$ was clearly overestimated at ranges from 0.5 to 1.46 m^{-1} , and underestimated above 1.46 m^{-1} ; for Lee's model, the $K_d(490)$ was significantly underestimated compared to the dataset used at ranges from 0.8 to 1 m^{-1} . These findings are also consistent with the results produced by Saulquin et al. (2013). Wang's $K_d(490)$ estimations were accurate for $K_d(490) < 0.8 \text{ m}^{-1}$. However, above 0.8 m^{-1} , the $K_d(490)$ become unstable and the retrieved uncertainty increases with the increase of $K_d(490)$ values. By comparison, Wang's model appears to be more effective than both Mueller and Lee's models when applied in the Bohai Sea. These findings imply that if the local bio-optical information is used to reinitialize the site-specific

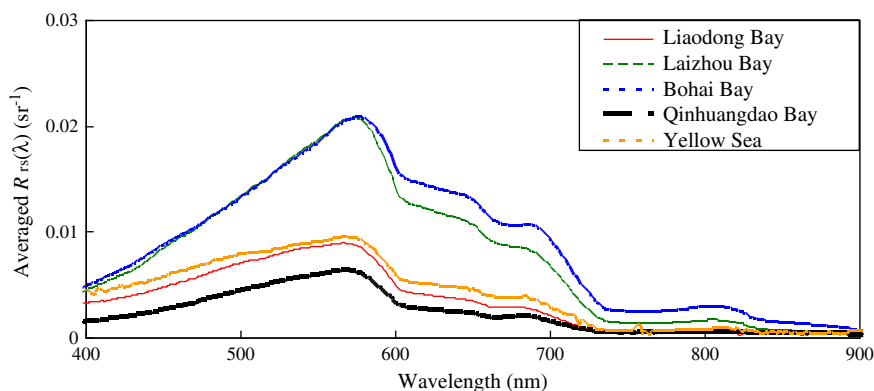


Fig. 6. Averaged remote sensing reflectance collected from Liaodong Bay, Yellow River Estuary, Bohai Bay, and Laizhou Bay.

Table 3
 $K_d(490)$ quantitative retrieval models.

Model	Adjusted model	R^2
Mueller	$1.446 - 2.289 \left[\frac{R_{rs}(490)}{R_{rs}(555)} \right]^{4.252}$	0.1367
Lee	$(1 + 0.005\theta_s)a(490) + 25.981[1.0 - 0.52e^{-1.8a(490)}]b_b(490)$	0.5087
Wang	$\frac{2.13 \times 10^{-4}}{R_{rs}(488)} + 0.8245 \frac{R_{rs}(667)}{R_{rs}(488)} + 3.657[18.877R_{rs}(667) - 0.0165] \left[1 - 0.52e^{-\frac{2.533 \times 10^{-3}}{R_{rs}(488)} - 9.81 \frac{R_{rs}(667)}{R_{rs}(488)}} \right]$	0.8841

coefficients of Wang's model, then Wang's model produces a good performance in deriving $K_d(490)$ from the Bohai Sea, which has a coefficient of determination (R^2) of 0.8841.

3.3.2. SSA model calibration

The band tuning technique was used to determine the optimal positions for λ_i and λ_j in the SSA model, i.e. in order to determine the optimal positions and constant coefficients of the SSA model, λ_i and λ_j were determined by bands tuning from 412 to 869 nm using MODIS spectral bands. The recursive procedures proposed by Chen and Quan (2013) were used to identify the best fit function of Eq. (16). RMS values, as shown in Eq. (2), were used to determine the optimal bands and constant coefficients of the SSA model. The optimal SSA model must have the minimum RMS value. For the purpose of prediction, statistical regression relationships were established between the SSA model and in situ $K_d(490)$ taken in September 2005 (Fig. 2A and B). Based on 84 field samples, the semi-analytical model shown in Fig. 8 was proposed as the optimal SSA model in quantifying $K_d(490)$ from the Bohai Sea. It was found that the optimal bands of the SSA model were $\lambda_i = 490$ nm and $\lambda_j = 667$ nm, and that the corresponding correlation coefficient between the retrieved $K_d(490)$ and in situ measurements was 0.941. The results indicate that the SSA model is an effective predictor in deriving $K_d(490)$ with MODIS spectral bands in the turbid coastal waters of the Bohai Sea.

3.3.3. Model evaluation and comparison

In this section of the paper, an evaluation of Mueller, Lee and Wang's models and the SSA model with MODIS spectral bands is presented. The evaluation was based on the comparison of the $K_d(490)$ predicted by these algorithms, with $K_d(490)$ measured analytically for the independent dataset collected from the Bohai Sea in June 2005 (Fig. 2C and D). Comparisons of the measured and predicted estimates of $K_d(490)$ by Mueller, Lee and Wang's models and the SSA model with MODIS spectral bands are presented in Fig. 9A. It was found that the SSA model ($RMS = 22.85\%$) had a superior performance in comparison to Mueller ($RMS = 107.4\%$), Lee ($RMS = 76.51\%$) and Wang's ($RMS = 55.76\%$) models. Using the SSA model for retrieving $K_d(490)$ in the Bohai Sea

decreased by 84.55% RMS from Mueller's model, 53.66% RMS from Lee's model, and 32.91% RMS from Wang's model. It appears that only the SSA model and Wang's model may be used to retrieve $K_d(490)$ from turbid Case II waters, but in these waters the SSA model is more effective than Wang's model.

The relationship between RE and $K_d(490)$ was also presented to demonstrate the ability of these four models in estimating $K_d(490)$ (Fig. 9B). It was found that the RE value decreases with increasing $K_d(490)$ values, but there is no statistically significant relationship between the two. For the range of $K_d(490)$ from 0.18 to 3.65 m^{-1} , the RE of $K_d(490)$ predicted by the SSA model was below 50%, with an average of 22.85% of the observed $K_d(490)$. The slopes of the linear relationships between SSA model-derived and in situ measured $K_d(490)$ was 1.01. Comparison of the SSA model to that of other three models indicates that the SSA model considerably reduces the RE value and outperforms Mueller, Lee and Wang's models, especially at a low $K_d(490)$ level ($<0.5 m^{-1}$). These findings imply that the SSA model does not require further optimization of spectral band positions and site-specific parameterization to accurately estimate $K_d(490)$ in water bodies with widely varying bio-optical characteristics taken in different seasons in the Bohai Sea.

3.4. $K_d(490)$ retrieval in the Bohai Sea

Due to the poor performance of Mueller and Lee's models in the Bohai Sea, only Wang's model and the SSA model are discussed in this section of the paper.

3.4.1. Atmospheric correction

Due to the fact that the majority of the Bohai Sea water falls into the Case II category (Cui et al., 2010), classical methods of atmospheric correction (NIR-based model) (Gordon & Wang, 1994; Gould & Arnone, 1997) may be violated in these waters. Wang, Son, and Shi (2009), Wang, Tang, and Shi (2007) proposed an atmospheric correction algorithm which uses the shortwave infrared bands (SWIR-based model) to improve ocean color products in turbid coastal waters measured by the MODIS. However, due to the considerably lower sensor signal-noise

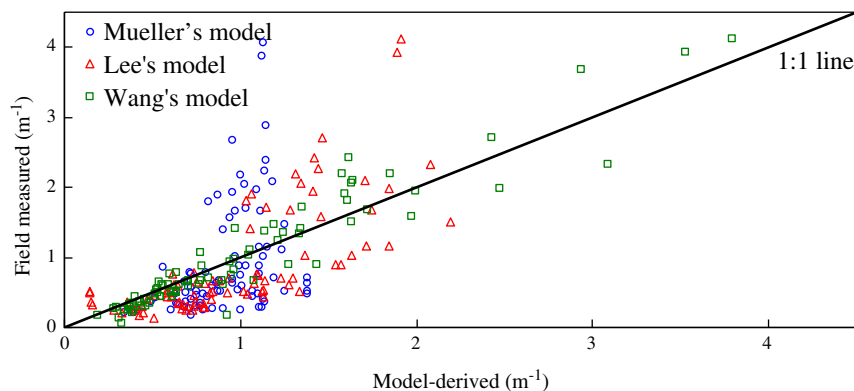


Fig. 7. Mueller's, Lee's, and Wang's models-derived plotting against field measured $K_d(490)$, 84 samples.

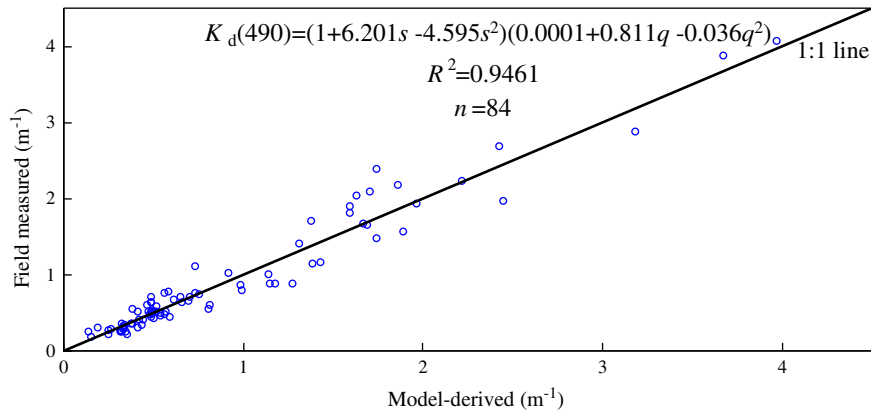


Fig. 8. SSA model-derived plotting against field measured $K_d(490)$ taken in Bohai Sea.

ratio (SNR) values for the MODIS short infrared bands, the SWIR-based model may fail in accurately supporting water-leaving products for the general application of ocean color remote sensing. In order to minimize the effects of this problem, an improved SWIR-based model (ISWIR-based) proposed by Chen, Cui, and Lin (2013) is used to retrieve remote sensing reflectance at desired wavelengths from MODIS data. In ISWIR-based model, the aerosol reflectance and aerosol type at near-infrared wavelengths are determined from SWIR wavelengths using a 5×5 box-based cross-calibration method initialized in clearly Case I waters, and then the NIR-based atmospheric correction model is used to determine the remote sensing reflectance from satellite-observed reflectance. The specific formulas of average values

of $\rho_{as}(869)$ and $\varepsilon(748,869)$ for a 5×5 box can be written as follows (Chen, Cui, et al., 2013):

$$\rho_{as}(869) = 30.333x^2 + 112.820y^2 - 134.374xy + 1.792x - 0.901y \quad (17)$$

$$\varepsilon(748,869) = \frac{\rho_{as}(748)}{\rho_{as}(869)} 4.544\gamma^2 - 8.169\gamma + 4.616 \quad (18)$$

$$\gamma = \frac{20.889x^2 + 114.825y^2 - 123.586xy + 2.058x - 1.283y}{\rho_{as}(869)} \quad (19)$$

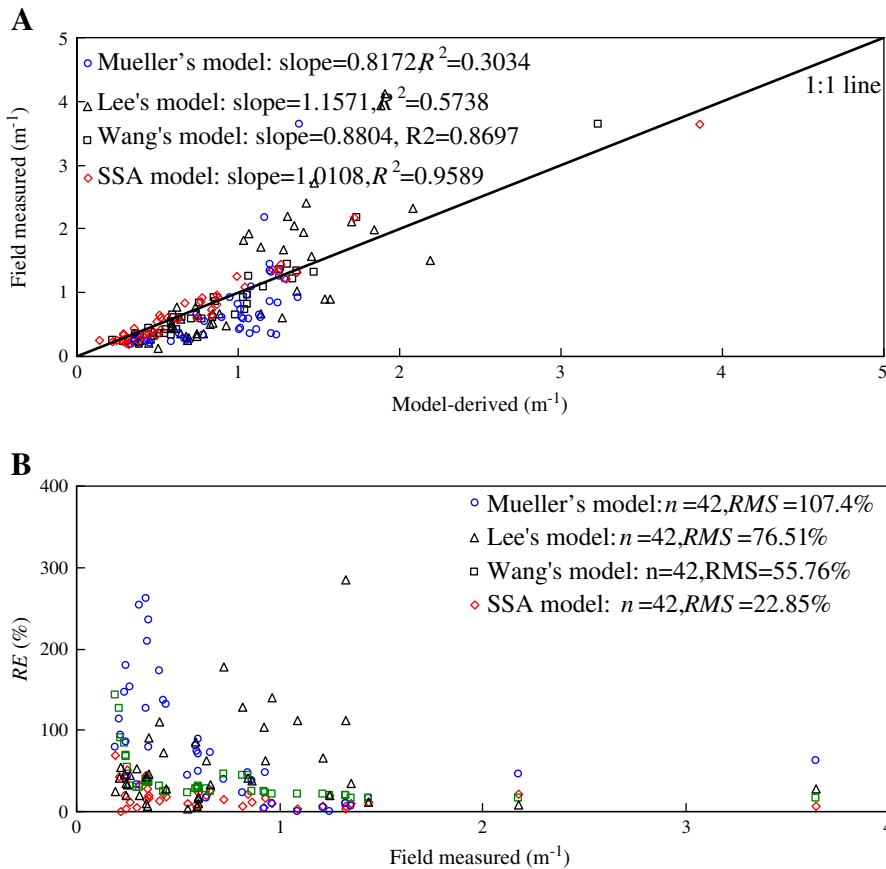


Fig. 9. Accuracy evaluation results of Mueller's, Lee's, Wang's, and SSA models. (A) Model-derived plotting against field measured $K_d(490)$; and (B) the relative error of $K_d(490)$ prediction plotted versus measured $K_d(490)$.

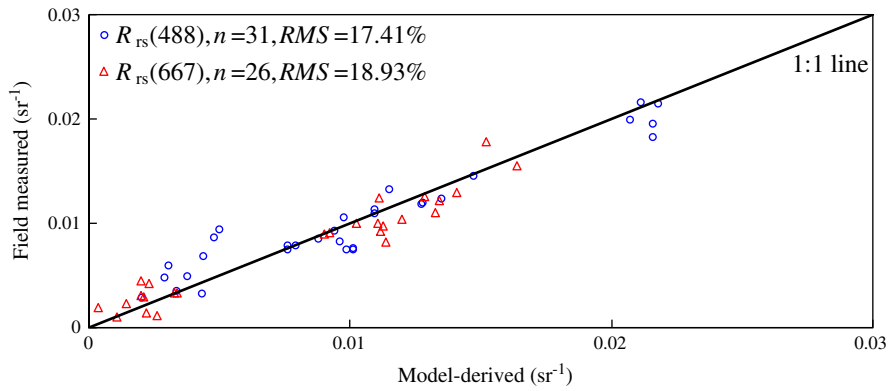


Fig. 10. ISWIR-based model-derived plotted against field measured remote sensing reflectance.

where x and y refer to $\rho_{as}(1240)$ and $\rho_{as}(2130)$, respectively, and ρ_{as} refers to the aerosol scattering reflectance.

In order to evaluate the performance of the ISWIR-based model in remote sensing reflectance deriving, four MODIS images, taken over the Bohai Sea, were selected for use in the experiments. The ISWIR-based model described by Chen, Cui, et al. (2013) was then used to process these images. The accuracy of the atmospheric correction model was evaluated through comparison between the retrieved and the observed $R_{rs}(488)$ and $R_{rs}(667)$. The observation stations within a ± 3 hour time window of satellite overpass and measurement were selected. The atmospheric conditions within this ± 3 hour period are reasonably stable (Bailey & Werdell, 2006). For the data match-up analysis, the procedure proposed by Bailey and Werdell (2006) was used to produce the satellite data for comparison with in situ measurements. Briefly, for a given MODIS-derived water-leaving reflectance, pixels with a 3×3 box centered at the locations of the in situ measurements were extracted, and the retrievals of the 3×3 pixels were averaged for the validation. The results (Fig. 10) indicate that the ISWIR-based model provides strong performance in remote sensing retrievals. Using ISWIR-based model produces 17.41 and 18.93% when deriving $R_{rs}(488)$ and $R_{rs}(667)$, respectively.

3.4.2. Accuracy evaluation

Three MODIS images were selected for use in the experiments. These are MODIS images which were scanned on September 1, 18, and 22, 2005, over the Bohai Sea. The accuracy of the $K_d(490)$ retrievals model was evaluated through comparison between the retrieved and the observed $K_d(490)$. The procedure of Bailey and Werdell (2006) mentioned in the previous section was used to produce the satellite-derived $K_d(490)$ for data match-up analysis. Fig. 11 shows the satellite-derived plotted against the field measured $K_d(490)$ within a ± 3 hour period as the satellite passed over the Bohai Sea, indicating that both Wang's

model and the SSA model provided strong performances in deriving $K_d(490)$ from the Bohai Sea, but the SSA model produces a superior performance to Wang's model. By comparison, use of the SSA model in estimating $K_d(490)$ in the Bohai Sea decreases the uncertainty of estimation by 12.1% RMS values from Wang's model. Therefore, both Wang's model and the SSA model are acceptable for accurately retrieving $K_d(490)$ from turbid coastal waters, but the SSA model may be more effective than Wang's model.

3.4.3. Overall comparison

Providing an overall SSA model performance evaluation in a reference to Wang's model, Fig. 12A shows $K_d(490)$ in the Bohai Sea on September 18, 2005, derived from the atmospheric correction results of the SWIR-based model. It was found that both models produced similar distribution trend of $K_d(490)$ in the Bohai Sea, i.e. the $K_d(490)$ from the seashore in all directions had the sharpest decline to the Central Bohai Sea. From the perspective of physical oceanography, the Bohai Sea is a semi-enclosed sea which has a free connection with the open sea. Several large rivers pour into the Bohai Sea, including the Yellow River, carrying a large amount of organic and inorganic suspended matter, with an average annual run-off of $5.8 \times 10^{10} \text{ m}^3$ and an annual sediment transport of 1.1×10^9 tons (Chen, Quan, et al., 2013). Cui et al. (2010) showed that, due to the fact that larger rivers pour into the Bohai Sea, the phytoplankton concentration and suspended sediment concentration along the coast are higher than in the Central Bohai Sea. As a result, the apparent offshore decreases of the $K_d(490)$ indicate that waters along the offshore are more turbid than those of the Central Bohai Sea.

However, Fig. 12A indicates several clear differences between the $K_d(490)$ derived from SSA model and that derived from Wang's model. For example, the former is much higher than the latter. This phenomenon is especially significant along the offshore, where the $K_d(490)$ derived by the SSA model is averaged 0.28 m^{-1} larger than that derived by Wang's

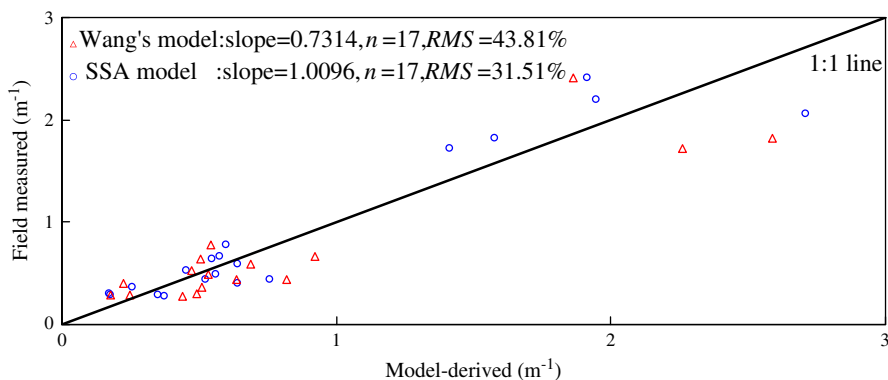


Fig. 11. Satellite-derived plotted against model-derived $K_d(490)$ in Bohai Sea.

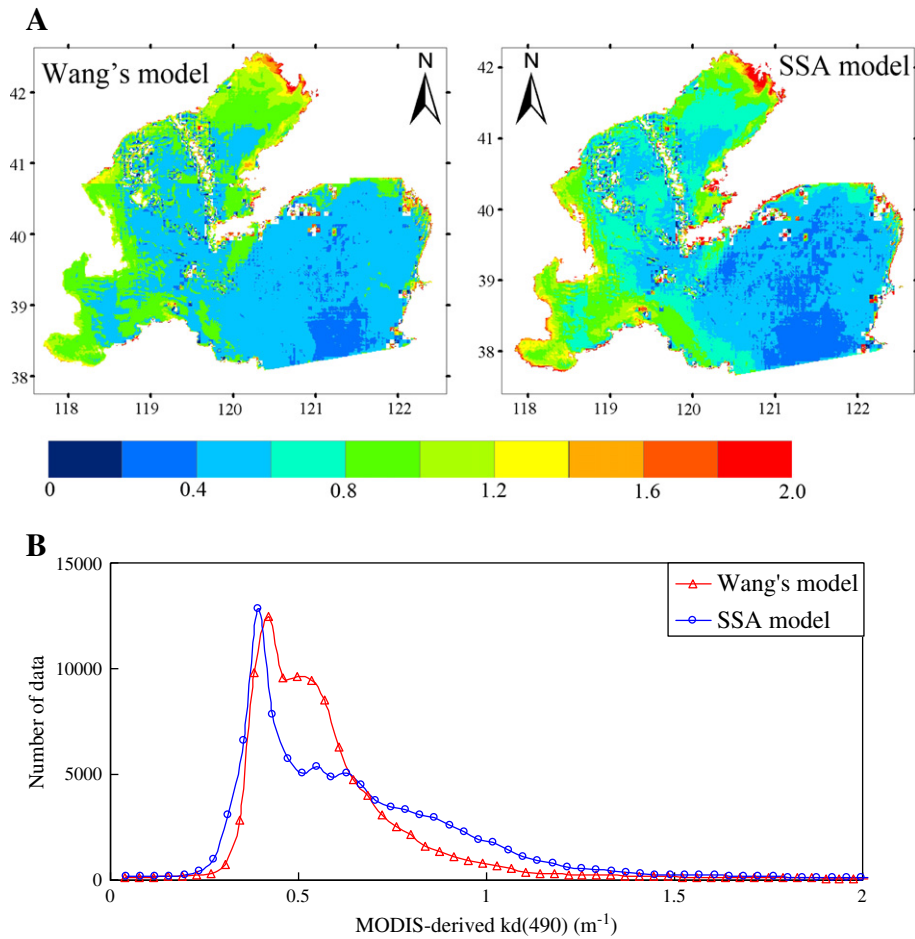


Fig. 12. Overall comparisons in $K_d(490)$ processed using Wang's and SSA models. (A) Wang's and SSA models-derived $K_d(490)$ in Bohai Sea on September 18, 2005; and (B) comparisons of histogram results in $K_d(490)$ processed using Wang's and SSA models.

model. Judging data quality by their coverage, spatial continuity, negative values, and image smoothness, it appears that the SSA model produces a superior performance in comparison to Wang's model in deriving $K_d(490)$ from MODIS data. Fig. 12B shows a histogram comparing the $K_d(490)$ results derived from the two models, indicating that the peak values for the $K_d(490)$ derived by the SSA model are slightly lower than those derived by Wang's model. For $K_d(490) < 0.39 \text{ m}^{-1}$ and $K_d(490) > 0.63 \text{ m}^{-1}$, the number points of the $K_d(490)$ values derived by the SSA model are much higher than those derived by Wang's model, while the number points of $K_d(490)$ processed by the SSA

model are much lower than Wang's results for $0.39 \text{ m}^{-1} < K_d(490) < 0.63 \text{ m}^{-1}$. These findings imply that the SSA model produces broad, flat frequency distributions of $K_d(490)$, in comparison to those produced using Wang's model.

4. Discussion

In order to evaluate the applicability of Mueller, Lee and Wang's models and the SSA model in determining $K_d(490)$ from productive and turbid coastal waters, these models were further validated using

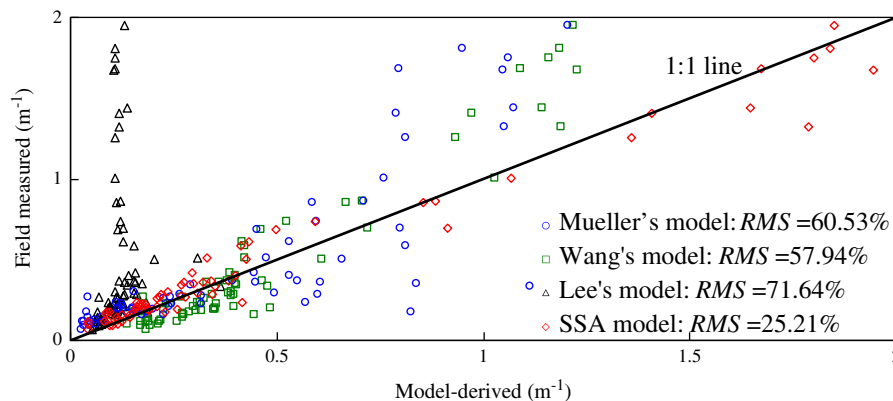


Fig. 13. Model-derived plotted against field measured $K_d(490)$ in Yellow Sea (69 samples).

independent dataset obtained over the Yellow Sea in April and September, 2003 (Fig. 2E and Table 1C). It was found (Fig. 13) that the SSA model does not require further optimization of spectral band positions or site-specific parameterization to accurately estimate $K_d(490)$ in water bodies with widely varying bio-optical characteristics, despite the fact that the bio-optical properties of the Yellow Sea differ from those used for model development (Table 1). For $K_d(490)$ ranging from 0.04 to 1.95 m^{-1} , the RE of the SSA model's $K_d(490)$ prediction did not exceed 56%, and had an average of 25.21% of the observed $K_d(490)$. Moreover, Fig. 13 indicates that Mueller, Lee and Wang's models produce an inferior performance in comparison to the SSA model, but Wang's model is more suitable than both Mueller and Lee's models. By comparison, use of the SSA model in estimating $K_d(490)$ in the Yellow Sea decreases the uncertainty of estimation by 32.73, 35.32, and 46.43% RMS values, from Wang, Mueller and Lee's models, respectively.

Models based on blue/green remote sensing reflectance ratios usually produce poor performance in shelf seas and turbid coastal waters (Lee et al., 2005; Wang, Son, & Harding, 2009), despite the fact that these models may produce a strong performance in deriving $K_d(490)$ from opening oceanic waters (Mueller & Trees, 1997). The reason for this may be attributed not only to the fundamental product of the algorithm structure, but also to the variability in inter-relationship between inherent optical properties of different water types. The variations in numerator and denominator values of blue/green $R_{rs}(\lambda)$ ratio are largely determined by absorption due to phytoplankton, suspended particles, and colored dissolved organic matter (CDOM) in productive and/or turbid waters. In these waters, the CDOM and/or phytoplankton (both pigments and detritus) absorb light so strongly that there is basically little water-leaving signal at blue regions exiting from the water columns. As a result, small variations in optically active constituent may not change the reflectance at blue wavelengths in optically complex waters. Moreover, all optical instruments have calibration problems in blue and the performance of atmospheric correction is worse in the blue part of the spectrum (Gordon & Voss, 1999; Wang et al., 2007). Consequently, Mueller's model, which is based on the blue/green $R_{rs}(\lambda)$ ratio, may fail in optically complex waters, due to the fact that the $K_d(490)$ is not the simple function of spectral ratio of backscattering or absorption coefficients at the blue-green wavelengths in these waters (Devlin et al., 2009; Gordon, 1989; Schaeffer et al., 2011).

The choice of the reference wavelength is arbitrary, given the fact that backscattering due to suspended particles at other wavelengths may be derived analytically (Zawada et al., 2007). Therefore, the spectral ratio of the backscattering coefficients will be only weakly dependent on TMS concentration, and the spectral ratio of $a(\lambda) + b_b(\lambda)$ due to optically active constituents may be analytically retrieved from the spectral ratio of $R_{rs}(\lambda)$. In open oceanic waters, $b_b(\lambda)$ is usually much smaller than $a(\lambda)$ and thus may be safely removed from $a(\lambda) + b_b(\lambda)$ (Carder et al., 2003). Therefore, the $a(\lambda)$ may be accurately derived using the QAA model in open oceanic waters. However, the QAA model may produce poor performance when deriving $a(\lambda)$ from productive and/or turbid waters, due to the fact that $b_b(\lambda)$ cannot be neglected in these waters (Chen & Quan, 2013; Chen, Quan, et al., 2013; Tassan, 1997). As a consequence, Lee's model may be violated in productive and/or turbid waters, due to the fact that this model is a QAA model-derived $a(\lambda)$ -based model. As a result, Lee's model produces a poor performance when deriving $K_d(490)$ from the Bohai Sea, as the majority of the its water falls into the productive and turbid Case II category (Cui et al., 2010). In addition to these observations, another limitation of Lee's model is that the absorption coefficient is calculated using an empirical model. It is well known that empirical models are generally only appropriate for application to waters with optical characteristics similar to those used in the development of the model (Matthews, 2011).

Wang's model appears to produce a superior performance to both Mueller and Lee's models. In Wang's model, the backscattering

coefficient is derived using the conceptual model of $b_b(490) \propto R(670)$. It is well known (Gordon & Clark, 1981) that $R(670)$ is not only related to $b_b(\lambda)$ but also related to $a(\lambda) + b_b(\lambda)$. In Wang's model, the influence of $a(\lambda) + b_b(\lambda)$ on $R(670)$ is ignored, and instead approximated using a constant H . This may be true when the bio-optical conditions are similar to those used during the development of Wang's model, the model may be violated in some complex cases. For example, the $K_d(490)$ may be underestimated by Wang's model when the waters are more turbid than those used for model development, due to the fact that $a(\lambda) + b_b(\lambda) > H$, and may be overestimated when the waters are less turbid than those used for model construction, as $a(\lambda) + b_b(\lambda) < H$. Fig. 13 also reveals that $K_d(490)$ is underestimated by Wang's model while $K_d(490) > 0.5 \text{ m}^{-1}$, but is overestimated by Wang's model while $K_d(490) < 0.5 \text{ m}^{-1}$. These findings imply that the applicability of Wang's model may be limited in terms of application to China's coastal waters, where the bio-optical properties are very complex.

The calibration and validation datasets used in this study were taken from the Bohai Sea in June–September 2005 and from Yellow Sea in April and September 2003. Although the variations of optically active constituents are quite wide and therefore the SSA model should be applicable in quite a wide variety of coastal waters, it may be still insufficient to completely validate the accuracy of the SSA model in other waters with different bio-optical properties found throughout the world. Thus, it is concluded that the SSA model should be used for estimating $K_d(490)$ in productive and turbid coastal waters, despite the fact that it may be essential to optimize the site-specific parameterization of the SSA model for the given aquatic bio-optical conditions as necessary. Other researchers have also suggested calibration and validation of the algorithms based on more in situ measurements of waters with different optical properties.

5. Summary

Accurate assessment of $K_d(490)$ in turbid waters by means of remote sensing is a challenging task, due to the optical complexity of productive and turbid coastal waters. There are several empirical and semi-analytical models for computing the $K_d(490)$. However, at present there is a compelling need for a diffuse attenuation model for turbid coastal waters, as the existing models are essentially applicable only for clear open ocean waters. The estimation accuracy of $K_d(490)$ appears to be greatly improved by Wang's model, but this model is only suitable for turbid waters which have bio-optical properties similar to those used for model development. As a consequence, an accurate $K_d(490)$ estimation model for turbid water remains under development. In ISWIR-based model, the aerosol reflectance and aerosol type at near-infrared wavelengths are determined from SWIR wavelengths using cross-calibration method initialized in clearly Case I waters.

The authors have evaluated the performances of three existing $K_d(490)$ models for the Bohai Sea and Yellow Sea, and further improved these for turbid coastal waters using an SSA model. The study results indicate that the SSA model produces a superior performance in deriving $K_d(490)$ from both the Bohai and Yellow Seas, in comparison to the existing three models selected for use in this study. It was also shown that use of the SSA model in estimating $K_d(490)$ decreased the uncertainty of estimation by >28.04% RMS values from the three existing models, in both the Bohai and Yellow Seas. The $K_d(490)$ were quantified from MODIS images after atmospheric correction using an ISWIR-based model, the remote sensing reflectance retrieved accuracy of which was 17.41 and 18.93% at 488 and 667 nm, respectively. After comparison between the satellite-derived and field measured $K_d(490)$, it was seen that the SSA model produces 31.51% RMS values in deriving $K_d(490)$ from MODIS data, which is 12.1% more accurate than Wang's model. These findings imply that the SSA model may be used for quantitative monitoring of $K_d(490)$ in coastal waters, provided that the atmospheric correction scheme is available for the MODIS imagery.

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Appendix A

The radiative transfer equation for radiance $L(z, \mu)$ in seawater which is stratified in plane layers may be written as follows (McCormick & Francisco, 1995):

$$\mu \frac{\partial L}{\partial z} + cL = b \int_{-1}^1 \beta(\mu, \mu') L d\mu' \quad (A1)$$

where z is the depth measured beneath the surface; μ is the cosine of the polar angle defined with respect to the downward direction; $c = a + b$ is the sum of the absorption and scattering coefficients; $\beta(\mu, \mu')$ is the scattering function for light scattered with direction μ from direction μ' ; and λ is the wavelength.

After integrating the radiative transfer equation for all directions of the light field, the following formula is obtained (Kylling & Stamnes, 1995):

$$\frac{d}{dz}(E_d - E_u) = -a(E_{0d} + E_{0u}). \quad (A2)$$

When the integration of Eq. (A1) is limited to the downward hemisphere, the result becomes the following:

$$\frac{dE_d}{dz} = \left(\frac{a}{\mu_d} + \frac{r_d b_b}{\mu_d} \right) E_d + \frac{r_u b_b}{\mu_u} E_u \quad (A3)$$

where the index d denotes downward and u upward; E_d and E_u are the downward and upward irradiances; E_{0d} and E_{0u} are the scalar mean the downward; μ_d and μ_u are the downward and upward average cosines, respectively; and r_d refers to the mean upward scattering coefficient of the downward traveling photons, while r_u refers to the mean downward scattering coefficient of the upward traveling photons, with both coefficients normalized with the backward scattering coefficient. The advantage of two-stream models compared to single scattering models is that they contain the effects of multiple scattering. Their disadvantage is that, in this study, they cannot predict the change in the shape of the radiance distribution with depth, due to the fact that the main focus of this study is K_d retrievals. Provided that the waters are sufficiently deep that reflected light from the bottom does not influence the irradiance of the upper layer, all irradiance measurements show that a first approximation of E_d should be of the following form (Gordon, 1989):

$$E_d(z) = E_d(0) \exp(K_d z). \quad (A4)$$

The irradiance ratio ρ is defined as the ratio between upward and downward irradiance (Morel & Gentili, 1996):

$$\rho = \frac{E_u(z)}{E_d(z)}. \quad (A5)$$

Substituting Eq. (A4) into Eq. (A2) yields a useful relation between ρ and K_d (Mobley, 1994):

$$K_d = \frac{\mu_u + \mu_d \rho}{\mu_u \mu_d (1 - \rho)} a. \quad (A6)$$

By inserting Eq. (A4) in Eq. (A3), the following may be observed (Aas, 1996):

$$\rho \approx \frac{\mu_u r_d}{\mu_u + \mu_d} s \quad (A7)$$

where s is the ratio of backscattering to absorption. When inserting Eq. (A6) into Eq. (A7), the result becomes the following:

$$K_d = \frac{1 + \mu_d p s}{\mu_d (1 - \mu_u p s)} a \quad (A8)$$

where p represents the ratio of r_d to the sum value of μ_u and μ_d .

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